Apache Spark and Object Stores —What you need to know

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Apache Hadoop committer & PMC
ASF member

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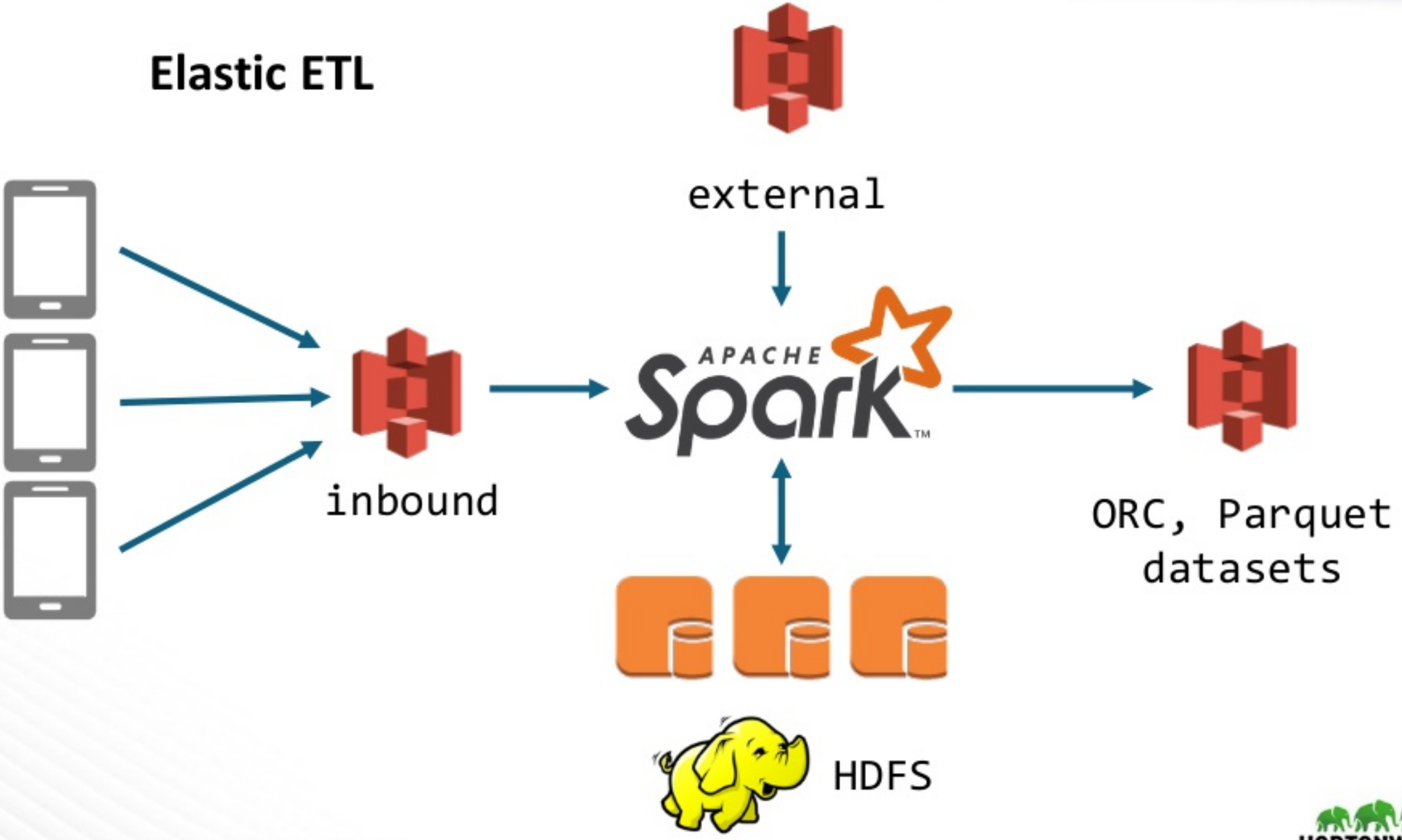
Tez Committer, PMC Member

Elastic ETL



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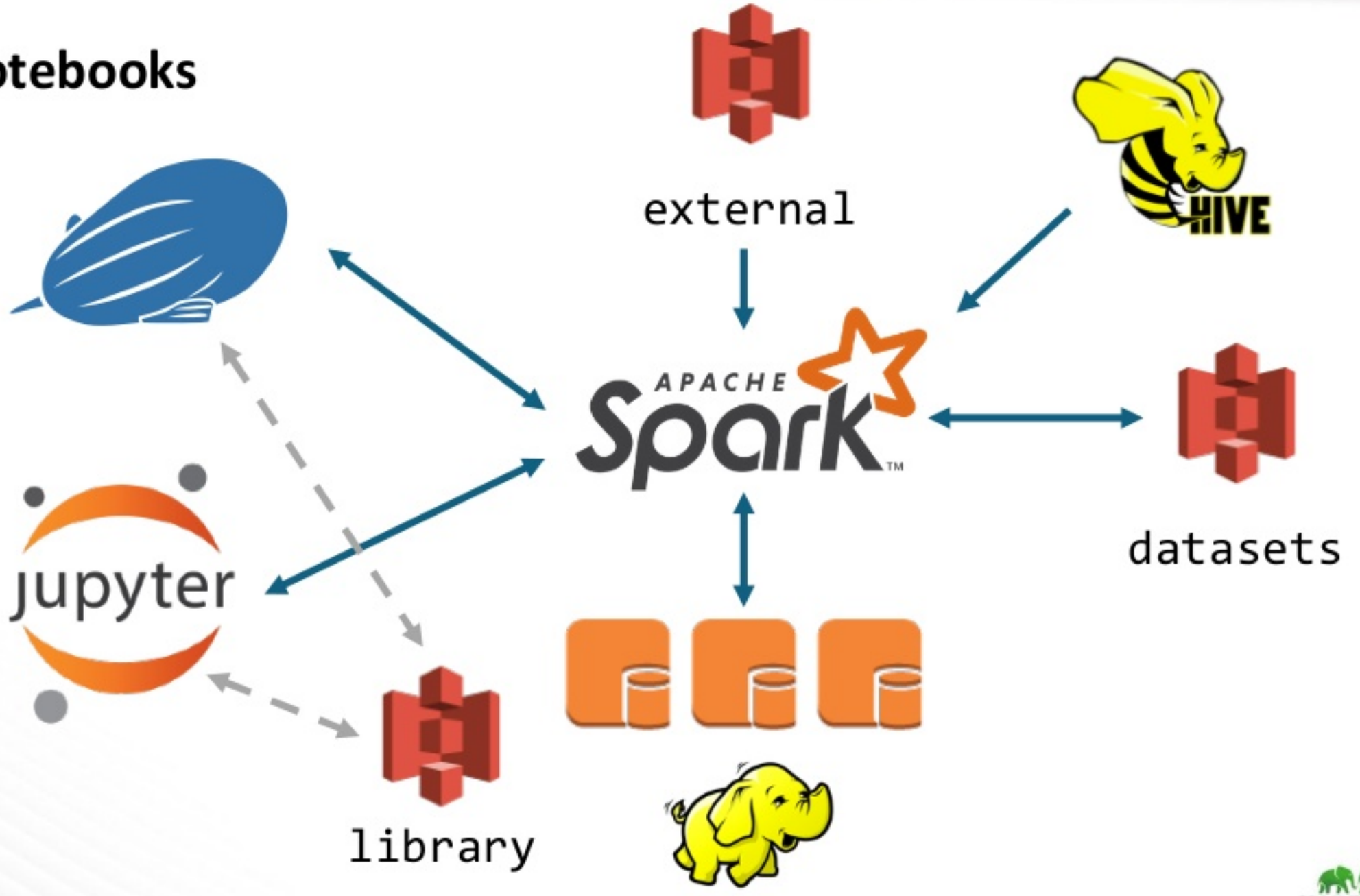
Notebooks



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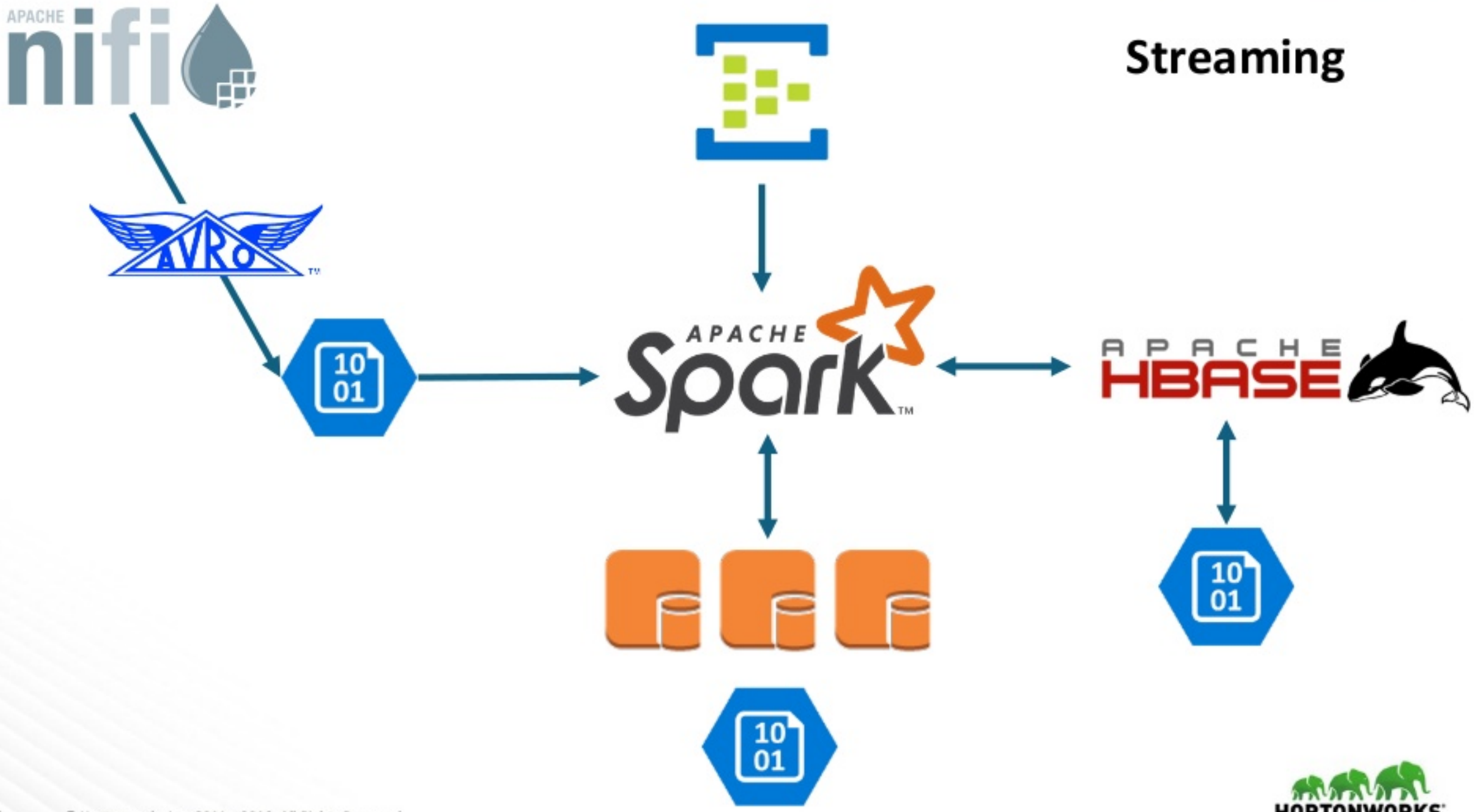
Streaming



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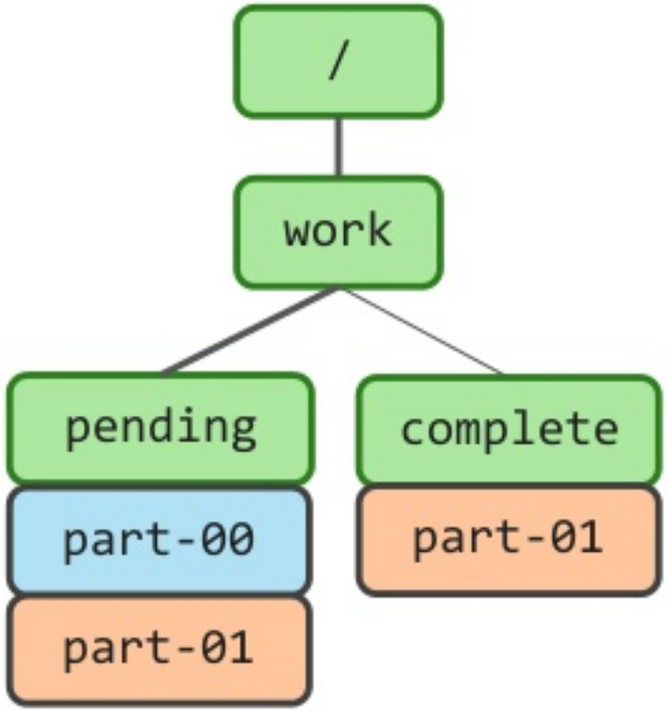
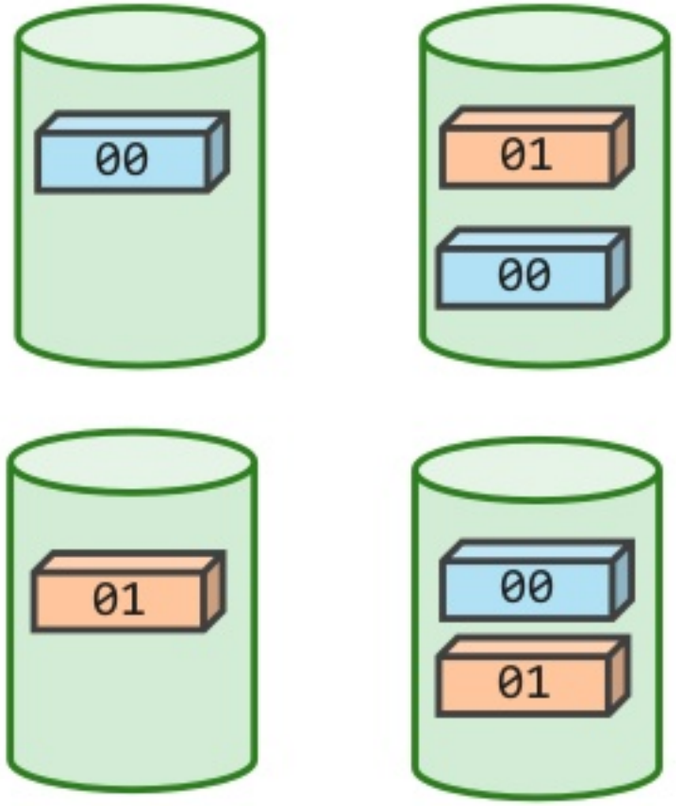
A Filesystem: Directories, Files → Data






```
rename("/work/pending/part-01", "/work/complete")
```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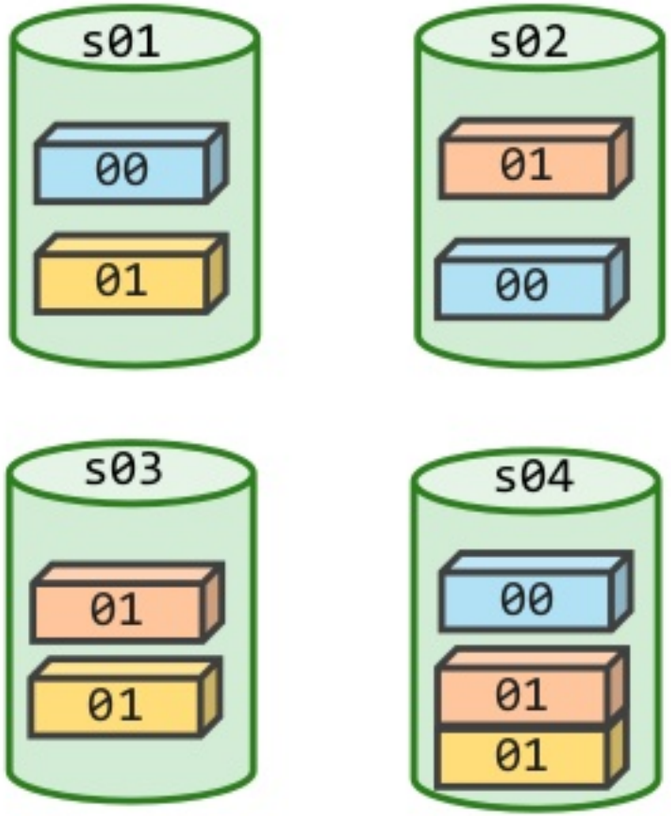
Object Store: hash(name)->blob

```
hash("/work/pending/part-00")
 ["s01", "s02", "s04"]

hash("/work/pending/part-01")
 ["s02", "s03", "s04"]

copy("/work/pending/part-01",
     "/work/complete/part01")

delete("/work/pending/part-01")
```



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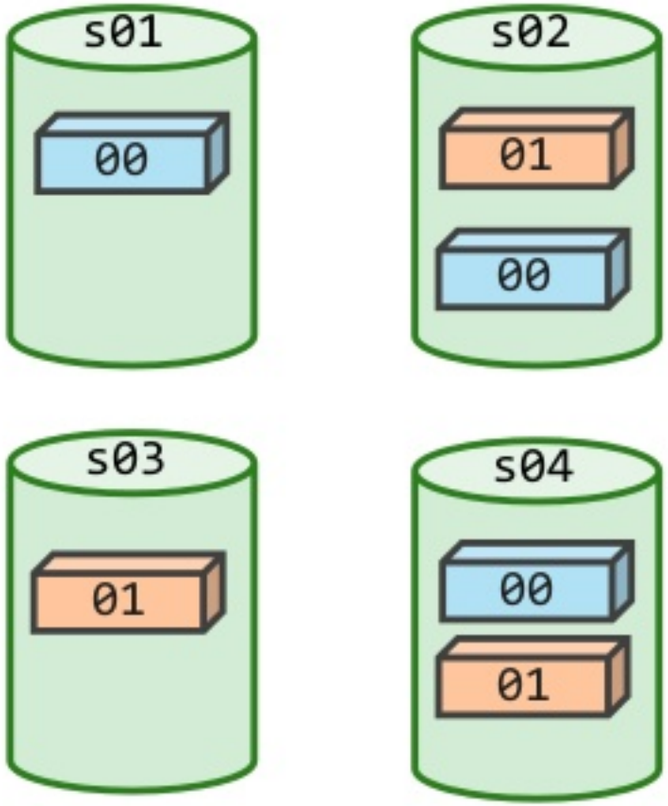
REST APIs

```
PUT /work/pending/part-01
... DATA ...

GET /work/pending/part-01
Content-Length: 1-8192

PUT /work/complete/part01
x-amz-copy-source: /work/pending/part-01

DELETE /work/pending/part-01

HEAD /work/complete/part-01

GET /?prefix=/work&delimiter=/
```



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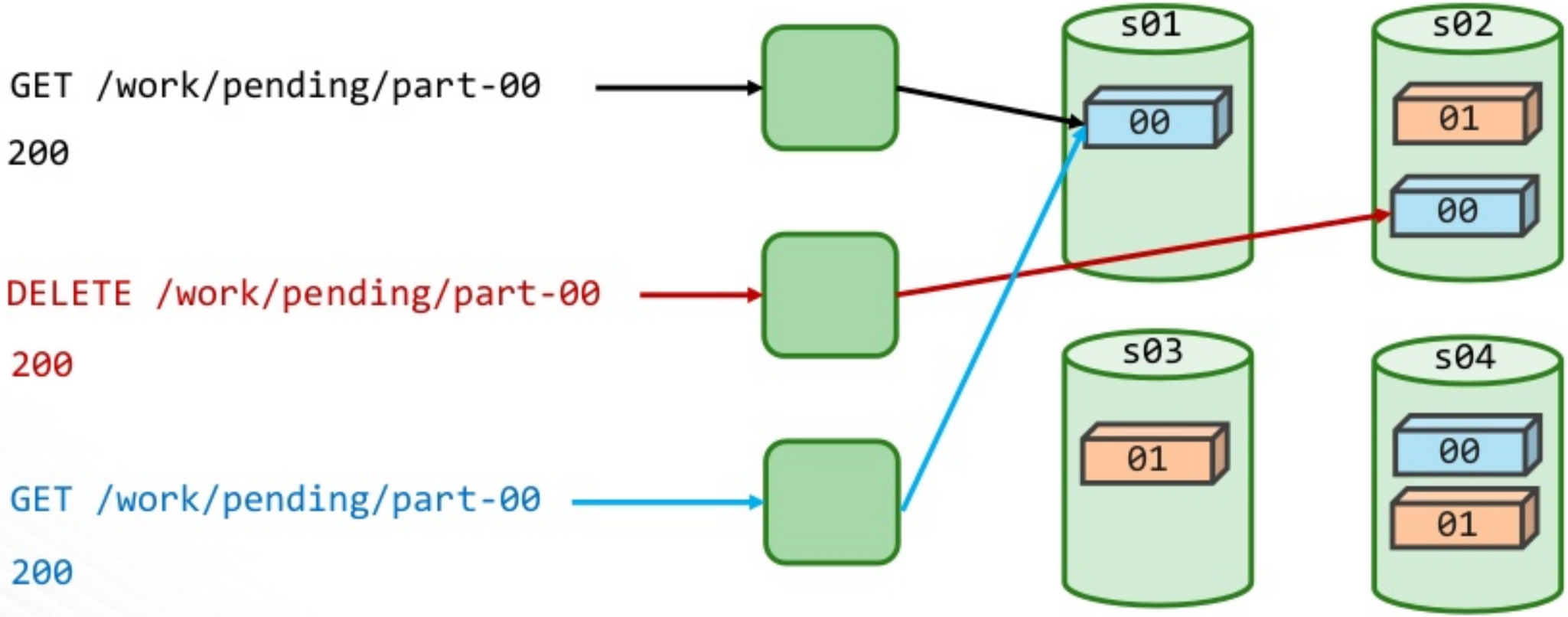
Often: Eventually Consistent



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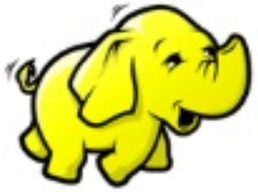
org.apache.hadoop.fs.FileSystem



hdfs

wasb

s3a

swift

adl

gs

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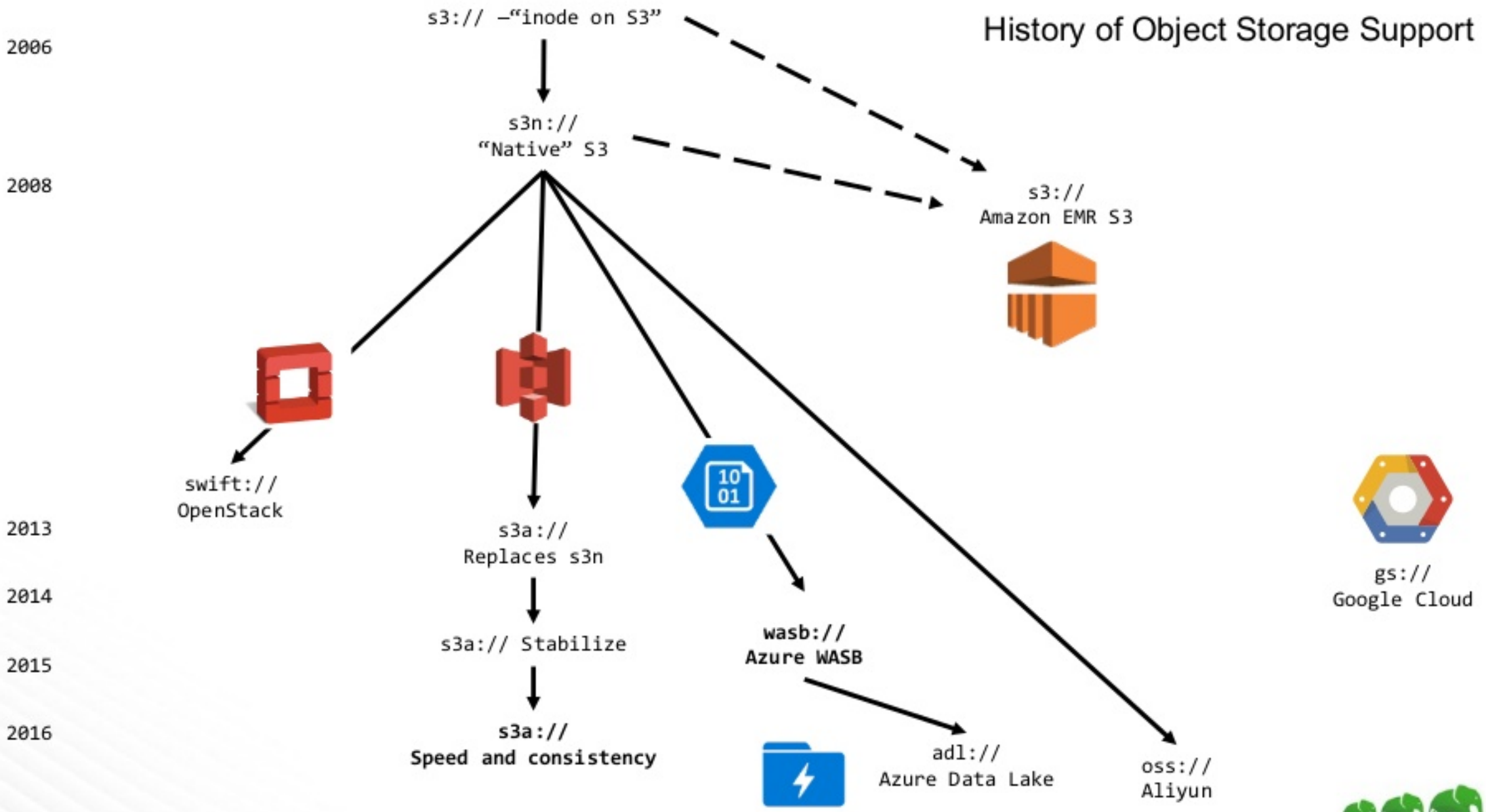
History of Object Storage Support



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Cloud Storage Connectors

Azure	WASB	• Strongly consistent • Good performance • Well-tested on applications (incl. HBase)
	ADL	• Strongly consistent • Tuned for big data analytics workloads
Amazon Web Services	S3A	• Eventually consistent - consistency work in progress by Hortonworks • Performance improvements in progress • Active development in Apache
	EMRFS	• Proprietary connector used in EMR • Optional strong consistency for a cost
Google Cloud Platform	GCS	• Multiple configurable consistency policies • Currently Google open source • Good performance • Could improve test coverage

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Four Challenges

1. Classpath
2. Credentials
3. Code
4. Commitment

Let's look At S3 and Azure

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Use S3A to work with S3

(EMR: use Amazon's s3://)

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Classpath: fix "No FileSystem for scheme: s3a"

hadoop-aws-2.7.x.jar

aws-java-sdk-1.7.4.jar
joda-time-2.9.3.jar
(jackson-*-2.6.5.jar)

Get Spark with Hadoop 2.7+ JARs

See SPARK-7481

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Credentials

`core-site.xml` or `spark-default.conf`

```
spark.hadoop.fs.s3a.access.key MY_ACCESS_KEY
spark.hadoop.fs.s3a.secret.key MY_SECRET_KEY
```

`spark-submit` automatically propagates Environment Variables

```
export AWS_ACCESS_KEY=MY_ACCESS_KEY
export AWS_SECRET_KEY=MY_SECRET_KEY
```

NEVER: share, check in to SCM, paste in bug reports…

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Authentication Failure: 403

```
com.amazonaws.services.s3.model.AmazonS3Exception:
 The request signature we calculated does not match
 the signature you provided.
 Check your key and signing method.
```

1. Check `joda-time.jar` & JVM version
2. Credentials wrong
3. Credentials not propagating
4. Local system clock (more likely on VMs)

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Code: Basic IO

```
// Read in public dataset
val lines = sc.textFile("s3a://landsat-pds/scene_list.gz")
val lineCount = lines.count()
```

```
// generate and write data
val numbers = sc.parallelize(1 to 10000)
numbers.saveAsTextFile("s3a://hwdev-stevel-demo/counts")
```

All you need is the URL

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Code: just use the URL of the object store

```
val csvdata = spark.read.options(Map(
    "header" -> "true",
    "inferSchema" -> "true",
    "mode" -> "FAILFAST"))
  .csv("s3a://landsat-pds/scene_list.gz")
```

...read time O(distance)

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DataFrames

```
val landsat = "s3a://stevel-demo/landsat"
csvData.write.parquet(landsat)

val landsatOrc = "s3a://stevel-demo/landsatOrc"
csvData.write.orc(landsatOrc)

val df = spark.read.parquet(landsat)
val orcDf = spark.read.parquet(landsatOrc)
```


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Finding dirty data with Spark SQL

```
val sqlDF = spark.sql(
  "SELECT id, acquisitionDate, cloudCover"
 + s" FROM parquet.`${landsat}`")

val negativeClouds = sqlDF.filter("cloudCover < 0")
negativeClouds.show()
```

* filter columns and data early
* whether/when to cache()?
* copy popular data to HDFS

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spark-default.conf

```
spark.sql.parquet.filterPushdown true
spark.sql.parquet.mergeSchema false
spark.hadoop.parquet.enable.summary-metadata false

spark.sql.orc.filterPushdown true
spark.sql.orc.splits.include.file.footer true
spark.sql.orc.cache.stripe.details.size 10000

spark.sql.hive.metastorePartitionPruning true
```


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Zeppelin

S3A-example

default

FINISHED

```
%pyspark
landsat = "s3a://landsat-pds/scene_list.gz"
landsat_rdd = sc.textFile(landsat)
print landsat_rdd.take(5)

```

```
[u'entityId,acquisitionDate,cloudCover,processingLevel,path,row,min_lat,min_lon,max_lat,max_lon,d
ownload_url', u'LC80101172015002LGN00,2015-01-02 15:49:05.571384,80.81,L1GT,10,117,-79.09923,-139
.66082,-77.7544,-125.09297,https://s3-us-west-2.amazonaws.com/landsat-pds/L8/010/117/LC8010117201
5002LGN00/index.html', u'LC80260392015002LGN00,2015-01-02 16:56:51.399666,90.84,L1GT,26,39,29.231
06,-97.48576,31.36421,-95.16029,https://s3-us-west-2.amazonaws.com/landsat-pds/L8/026/039/LC80260
392015002LGN00/index.html', u'LC82270742015002LGN00,2015-01-02 13:53:02.047000,83.44,L1GT,227,74,
-21.28598,-59.27736,-19.17398,-57.07423,https://s3-us-west-2.amazonaws.com/landsat-pds/L8/227/074
/LC82270742015002LGN00/index.html', u'LC82270732015002LGN00,2015-01-02 13:52:38.110317,52.29,L1T,
227,73,-19.84365,-58.93258,-17.73324,-56.74692,https://s3-us-west-2.amazonaws.com/landsat-pds/L8/
227/073/LC82270732015002LGN00/index.html']
```

Took 29 sec. Last updated by anonymous at October 21 2016, 9:34:43 PM.

READY

Notebooks? Classpath & Credentials

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The Commitment Problem

- `rename()` used for atomic commitment transaction
- time to copy() + delete() proportional to data * files
- S3: 6+ MB/s
- Azure: a lot faster —*usually*

```
spark.speculation false
spark.hadoop.mapreduce.fileoutputcommitter.algorithm.version 2
spark.hadoop.mapreduce.fileoutputcommitter.cleanup.skipped true
```


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What about Direct Output Committers?

[SPARK-10063] Remove Direc

https://issues.apache.org/jira/browse/SPARK-10063

Dashboards Projects More Create Search

Spark / SPARK-10063

Remove DirectParquetOutputCommitter

Edit | Comment | Agile Board | More | Close Issue | Reopen Issue

Details

Type:	Bug	Status:	RESOLVED
Priority:	Critical	Resolution:	Fixed
Affects Version/s:	None	Fix Version/s:	2.0.0
Component/s:	SQL		
Labels:	None		
Target Version/s:	2.0.0		

People

Assignee: Reynold Xin

Reporter: Yin Huai

Votes: 0 Vote for this issue

Watchers: 14 Stop watching this is

Description

When we use DirectParquetOutputCommitter on S3 and speculation is enabled, there is a chance that we can loss data.

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Recent S3A Performance (Hadoop 2.8, HDP 2.5, CDH 5.9 (?))

```
// forward seek by skipping stream
spark.hadoop.fs.s3a.readahead.range 157810688

// faster backward seek for ORC and Parquet input
spark.hadoop.fs.s3a.experimental.input.fadvise random

// PUT blocks in separate threads
spark.hadoop.fs.s3a.fast.output.enabled true
```


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Azure Storage: wasb://

A full substitute for HDFS

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Classpath: fix “No FileSystem for scheme: wasb”

`wasb://` : Consistent, with very fast rename (hence: commits)

```
hadoop-azure-2.7.x.jar
azure-storage-2.2.0.jar
+ (jackson-core; http-components, hadoop-common)
```


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Credentials: `core-site.xml` / `spark-default.conf`

```
<property>
 <name>fs.azure.account.key.example.blob.core.windows.net</name>
 <value>0c0d44ac83ad7f94b0997b36e6e9a25b49a1394c</value>
</property>

spark.hadoop.fs.azure.account.key.example.blob.core.windows.net
  0c0d44ac83ad7f94b0997b36e6e9a25b49a1394c
```

`wasb://demo@example.blob.core.windows.net`

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Example: Azure Storage and Streaming

```
val streaming = new StreamingContext(sparkConf,Seconds(10))
val azure = "wasb://demo@example.blob.core.windows.net/in"
val lines = streaming.textFileStream(azure)
val matches = lines.map(line => {
    println(line)
    line
  })
matches.print()
streaming.start()
```

* PUT into the streaming directory
* keep the dir clean
* size window for slow scans

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Not Covered

- Partitioning/directory layout
- Infrastructure Throttling
- Optimal path names
- Error handling
- Metrics

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Summary

- Object Stores look just like any other URL
- …but do need classpath and configuration
- Issues: performance, commitment
- *Use Hadoop 2.7+ JARs*
- *Tune to reduce I/O*
- *Keep those credentials secret!*

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APACHE:
BIG_DATA
EUROPE
November 14 - 16, 2016
Melia Sevilla, Seville, Spain
#ApacheBigData

Backup Slides

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Dependencies in Hadoop 2.8

hadoop-aws-2.8.x.jar

```
aws-java-sdk-core-1.10.6.jar
aws-java-sdk-kms-1.10.6.jar
aws-java-sdk-s3-1.10.6.jar
joda-time-2.9.3.jar
(jackson-*-2.6.5.jar)
```

hadoop-aws-2.8.x.jar

```
azure-storage-4.2.0.jar
```


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S3 Server-Side Encryption

- Encryption of data at rest at S3
- Supports the SSE-S3 option: each object encrypted by a unique key using AES-256 cipher
- Now covered in S3A automated test suites
- Support for additional options under development (SSE-KMS and SSE-C)

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Advanced authentication

```
<property>
  <name>fs.s3a.aws.credentials.provider</name>
  <value>
   org.apache.hadoop.fs.s3a.SimpleAWSCredentialsProvider,
   org.apache.hadoop.fs.s3a.TemporaryAWSCredentialsProvider,
   com.amazonaws.auth.EnvironmentVariableCredentialsProvider,
   com.amazonaws.auth.InstanceProfileCredentialsProvider,
   org.apache.hadoop.fs.s3a.AnonymousAWSCredentialsProvider
  </value>
</property>
```

+encrypted credentials in JECKS files on HDFS

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What Next? Performance and integration

[HADOOP-13525] Optimize u

https://issues.apache.org/jira/browse/HADOOP-13525

Dashboards · Projects · Issues · Service Desk · Agile · Create

Hadoop Common / HADOOP-13204 Über-jira: S3a phase III: scale and tuning / HADOOP-13525

Optimize uses of FS operations in the ASF analysis frameworks and libraries

Edit | Comment | Assign | More | Submit Patch | Start Progress | Resolve Issue | Export

Details

Type:	Sub-task	Status:	OPEN
Priority:	Major	Resolution:	Unresolved
Affects Version/s:	2.7.2	Fix Version/s:	None
Component/s:	fs, fs/s3		
Labels:	None		

People

Assignee: Steve Loughran
Reporter: Steve Loughran
Votes: 0
Watchers: 3 Stop watching this issue

Dates

Created: 20/Aug/16 12:33
Updated: 20/Aug/16 12:43

Agile

View on Board

Description

Review uses of the FS APIs in applications using the Hadoop FS API to access filesystems; identify suboptimal uses and tune them for better performance against HDFS and object stores

- Assume arbitrary Hadoop 2.x releases: make no changes which are known to make operations on older versions of Hadoop slower
- Do propose those changes which deliver speedups in later versions of Hadoop, while not impacting older versions, or risk of causing scalability problems.
- Add more tests, especially scalable ones which also display metrics.
- Use standard benchmarks and optimization tools to identify hotspots.
- Use FS behaviour as verified in the FS contract tests as evidence that filesystems correctly implement the Hadoop FS APIs. If a use of an API call is made which hints at the expectation of different/untested behaviours, leave alone and add new tests to the Hadoop FS contract to determine cross-FS semantics.
- Focus on the startup, split calculation and directory scanning operations: the ones which slow down entire queries.
- Eliminate use of `isDirectory()`, `getLength()`, `exists()` if a followon operation (`getStatus()`, `delete()`, ... makes the use redundant.
- Assume that `FileStatus` entries are not cached; the cost of creating them is 1 RPC call against HDFS, 1+ HTTPS call against object stores.
- Locate calls to the listing operations, identify speedups, especially on recursive directory scans.
- Identify optimized seek patterns (backwards, sideways, forwards) and attempt to replicate them in default caching.
- Try to reuse the results of previous operations (e.g `FileStatus` instances) in follow-on calls.
- Commonly used file formats (e.g ORC) will have transitive benefits.
- Frameworks to use predicate pushdown where this delivers speedups
- Document test patterns, scenarios and implements

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Next Steps for all Object Stores

- Output Committers
  - Logical commit operation decoupled from rename (non-atomic and costly in object stores)
- Object Store Abstraction Layer
  - Avoid impedance mismatch with FileSystem API
  - Provide specific APIs for better integration with object stores: saving, listing, copying
- Ongoing Performance Improvement
- Consistency

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